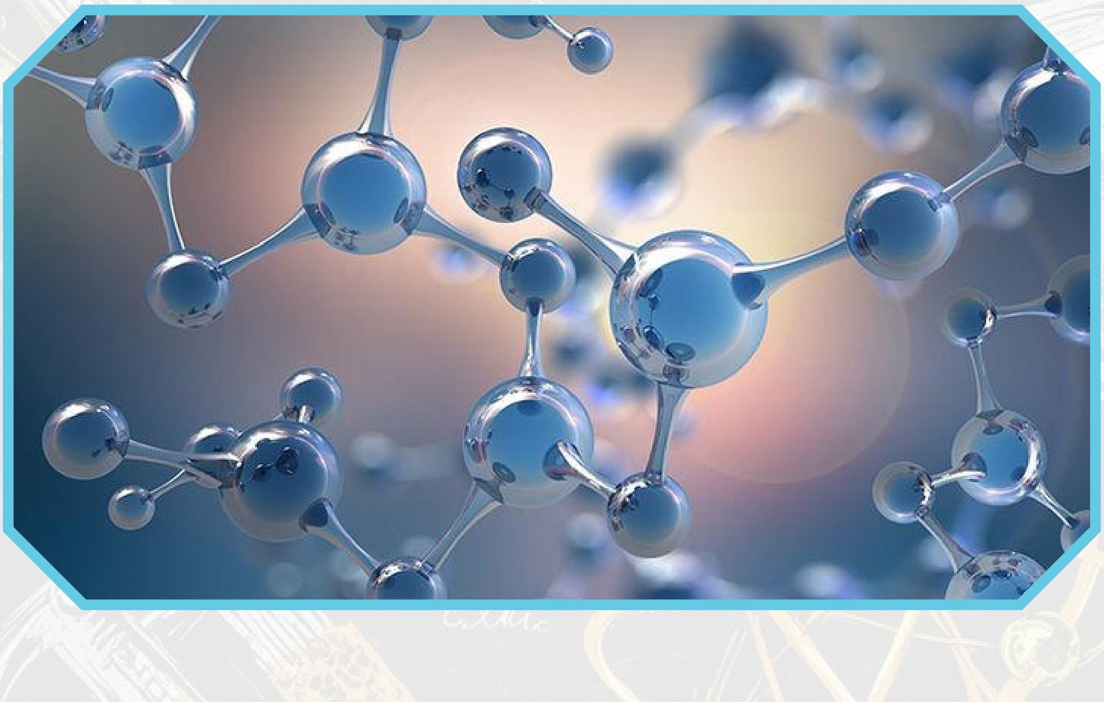
UNIT 27

NUCLEAR PHYSICS



Isotopes:
Two or more forms of the same element that contain equal numbers of protons but different numbers of neutrons in their nuclei, and hence they differ in relative atomic mass but not in chemical properties.
Isotopes are of two types:
• **Stable isotopes:**
They do not show any radioactivity and are stable. For example, $_7N^{14}$ and $_7N^{15}$.
• **Radioactive isotopes:**
o They are unstable and show radioactivity. Some of the radioactive isotopes are $_6C^{14}$ for carbon, and $_{19}K^{40}$ for potassium.

Every element on the periodic table naturally occurs in various forms having different masses like isotopic forms, although some may be more prevalent than the isotopes of hydrogen others. For instance:
• **Oxygen:**
About 99.76% of oxygen in nature is Oxygen-16 (with 8 neutrons). However, Oxygen-17 and Oxygen-18 are also found in much smaller quantities.

Uranium:
The naturally occurring isotopes of uranium are Uranium-238 (99.3%) and 235 Uranium-235 (0.7%). Uranium has 35 known isotopes. The most stable isotope of uranium is uranium-238, which has 146 neutrons.

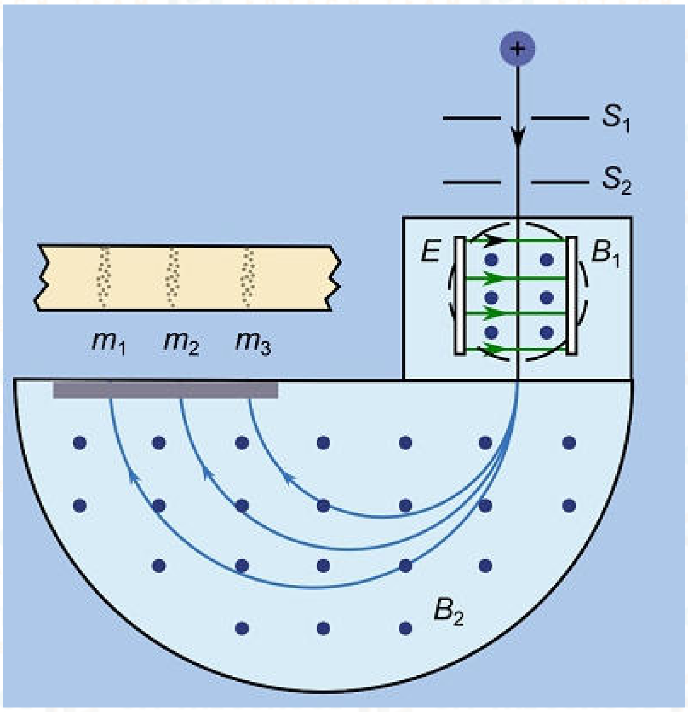
Hydrogen:
Hydrogen has only three isotopes: protium, deuterium, and tritium. Protium has no neutrons, deuterium has one neutron, and tritium has two neutrons.

Mass Spectrograph:
A mass spectrograph is a device that can be used to separate isotopes of an element based on their mass and it works by accelerating charged particles through a magnetic field. The particles are then deflected by the magnetic field, and the amount of deflection is determined by the mass of the particle.

Bainbridge Mass Spectrograph:
Bainbridge mass spectrograph is a type of mass spectrometer that uses a combination of electric and magnetic fields to separate ions according to their charge-to-mass ratio. It is named after its inventor, Kenneth T.
Bainbridge, who developed it in 1933.

Principle:
The fundamental principle behind the Bainbridge Mass Spectrograph is the application of magnetic and electric fields to separate and measure the masses of charged particles, typically ions. This separation is based on the principles of magnetic deflection and kinetic energy.

The Bainbridge mass spectrograph consists of three main components: an ion source, a velocity selector, and a magnetic analyzer.



• **Ion Source:** The ion source is where the ions are produced. The ions are typically produced by bombarding a sample with electrons, which knocks electrons out of the atoms, creating positively charged ions.

• **Velocity Selector:** The velocity selector is used to select ions of a particular velocity. The ions are passed through a region of electric and magnetic fields, which are adjusted so that only ions of a certain velocity can pass through.

• **Magnetic Analyzer:** The magnetic analyzer is used to separate the ions according to their charge-to-mass ratio. The ions are passed through a region of magnetic field, and the radius of their path is determined by their charge-to-mass ratio.

• The ions that are separated by the magnetic analyzer are then detected by a detector, such as a photographic plate or a computer.

A schematic diagram showing the construction of Bainbridge's mass spectrograph is shown in Figure, beam of positive ions produced in a discharge tube is collimated into a narrow beam by the two slits S_{61} and S_2. After emerging from the slit, the positive ions enter a velocity selector.
The velocity selector consists of two plates E and B, between which a steady **electric field is maintained in a direction at right angles to the ion beam.**
The electric field and the magnetic field of the velocity selector are so adjusted that the deflection produced by one is exactly equal and opposite to the deflection produced by the other so that there is no deflection for ions having a particular:

$$xe = B\,e\,v$$

$$\boldsymbol{v = x/\,B_1}$$

The ions were accelerated to a known kinetic energy, ensuring that all ions of different masses had the same kinetic energy. Positive ions entering the evacuated chamber are subjected to a perpendicular electromagnetic field of intensity B₂, causing them to follow curved paths. The curvature depends on their charge-to-mass ratio (e/m), with heavier ions curving less than lighter ones. This method is very accurate due to the linear mass scale. When a charged particle with mass m and charge e is accelerated through a potential difference V, it gains a velocity v, given by

$$\frac{1}{2}mv^2 = eV \quad or \quad v = \sqrt{\frac{2eV}{m}}$$

When this charged particle moving with a velocity v enters a magnetic field B in a direction perpendicular to the field, the force acting on it due to the field is Bev acting in a direction at right angles to the direction of motion of the charged particle and that of the magnetic field. The particle, therefore, moves along a circular path of radius r given by; Putting v in above:

$$Bev = \frac{mv^2}{r}$$

for mass m and putting value of v from equation $v = \sqrt{\frac{2eV}{m}}$),

$$m = \frac{Ber}{v} = \frac{Ber}{\sqrt{\frac{2eV}{m}}}$$

Squaring both sides and after re-arranging,

$$m = \left(\frac{er^2}{2V}\right)B^2$$

The mass of each ion reaching the detector depends on the value of B^2. By changing B and keeping other variables constant, ions of different masses can be directed into the detector.

This allows us to create a graph of detector readings versus B, identifying the masses and quantities of the ions present. Multiple peaks in the mass spectrum indicate different isotopes of the same element.

Radioactivity is the process by which an unstable atomic nucleus loses energy by radiation. A material containing unstable nuclei is considered radioactive.

There are two types of radioactivity:

1. Natural Radioactivity:

Natural radioactivity is the radioactivity that occurs naturally in the environment. It is caused by the decay of unstable isotopes of elements that are found in the Earth's crust, such as uranium, thorium, and potassium-40.

2. Induced Radioactivity:

Induced radioactivity is the radioactivity that is created by bombarding a stable material with radiation, such as neutrons or protons. This can be done in a nuclear reactor or particle accelerator.

The main difference between natural and induced radioactivity is the source of the radiation.

o In natural radioactivity, the radiation comes from the unstable nuclei of the atoms themselves.

o In induced radioactivity, the radiation comes from outside the atoms and is used to destabilize their nuclei.

Radioactive Decay Process:
The natural radioactivity observed was specific to heavy elements with atomic weights greater than about 206. Of the 2,500 identified types of atoms, around 90% are radioactive, meaning they change into other types of atoms over time. These unstable nuclides release energy and particles in much different ways of decay types to become stable nuclides.

Alpha Decay:
All atomic nuclei with a proton number (Z) greater than 83 and a mass number (A) greater than 209 undergo spontaneous transformation into lighter nuclei by emitting one or more alpha particles, which are equivalent to helium-4 nuclei.

$$^{A}_{Z}X \rightarrow ^{A-4}_{Z-2}Y + ^{4+}_{2}He$$

Where X is parent nuclei, Y is daughter nuclei, and He is alpha particle. Examples of alpha decays are:

$$^{238}_{92}U \rightarrow ^{234}_{90}Th + ^{4+}_{2}He$$

Beta Decay:
The beta-decay is a spontaneous process in which the mass number of the nucleus remains unchanged, but the atomic number changes by unity ($\Delta Z = +1$).
The change in atomic number due to emission of an electron, emission of positron or by capture of an orbital electron (K-capture).
If the daughter nucleus doesn't have the right ratio of neutrons and protons to stay stable, it can change through beta decay to become more stable.
There are three types of β-decay processes depending upon their mode of decay:
• β^- decay (electron emission)
• β^+ decay (positron emission) and
• electron capture.

β-Decays (Electron Emission):
General equation of electron emission is:

$$AX \rightarrow AY + e^-$$

Whenever, the number of neutrons is more than the number of protons, negative beta decay happens. In this process, a neutron is transformed into a proton:

$$1n \rightarrow 1p + e^-$$

Here is an example:

$$_{6}^{14}C \rightarrow {}_{6}^{14}N + e^-$$

This equation represents the decay of carbon-14 ($_{6}C^{14}$) into nitrogen-14 ($_{7}N^{14}$) with the emission of an electron.

β⁺-Decay (Positron Emission):
Also known as positive beta decay.
General equation of positron emission is:

$$AX \rightarrow AY + e^+$$

Whenever, the number of protons is more than the number of neutrons, positive beta decay happens. In this process, a proton is transformed into a neutron:

$$1p \rightarrow 1n + e+$$

Here is an example:

$${}_{6}^{11}C \rightarrow {}_{5}^{11}N + e+$$

This equation represents the decay of carbon-11 (${}_{6}C^{11}$) into boron-11 (${}_{5}B^{11}$) with the emission of a positron (e+).

Electron Capture:

In this process an electron from K-shell of the atom is captured by the nucleus to form a new nucleus and a photon is emitted:

$$AX + e- \rightarrow AY + h\nu$$

Gamma Decay:

During gamma decay, there is no change in mass number A or atomic number Z. An excited nucleus is denoted by an asterisk (*) after or over its usual symbol. Thus, 38S87 refers ${}_{38}S^{87}$ to in an excited state. The general equation for gamma decay is:

$$AX^{*} \rightarrow AX + \gamma$$

Here is an example of gamma decay:

$${}_{27}^{60}Co^{*} \rightarrow {}_{27}^{60}Co + \gamma$$

This equation represents the decay of an excited state of cobalt-60.